

MAT3032 ANSWERS TO EXERCISES 3.1

Exercises 3.1

1. $0_R r = 0_R = r 0_R$ for all $r \in R$ so $0_R \in Z(R)$, hence $Z(R)$ is non-empty. It is a subset of R by definition.

Let $a, b \in Z(R)$, so $ar = ra$, $br = rb$ for all $r \in R$.

Then $(a - b)r = ar - br = ra - rb = r(a - b)$, so $a - b \in Z(R)$

and $(ab)r = a(br) = a(rb) = (ar)b = (ra)b = r(ab)$, so $ab \in Z(R)$.

Hence $Z(R)$ is a subring of R .

2. The zero matrix is in S , so S is non-empty. It is a subset of $M_2(\mathbb{R})$ by definition.

Let $A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, $B = \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} \in S$.

Then $A - B = \begin{pmatrix} a - c & 0 \\ 0 & b - d \end{pmatrix} \in S$ and $AB = \begin{pmatrix} ac & 0 \\ 0 & bd \end{pmatrix} \in S$,

so S is a subring of $M_n(\mathbb{R})$.

We have $\phi(a, b) + \phi(c, d) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} + \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} a + c & 0 \\ 0 & b + d \end{pmatrix}$

$= \phi(a + c, b + d) = \phi((a, b) + (c, d))$

and $\phi(a, b)\phi(c, d) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} ac & 0 \\ 0 & bd \end{pmatrix} = \phi(ac, bd) = \phi((a, b)(c, d))$

so ϕ is a ring homomorphism. Clearly $\phi(\mathbb{R} \times \mathbb{R}) = S$, so ϕ is surjective.

$\text{Ker } \phi = \{(0, 0)\}$ so ϕ is injective. Hence ϕ is a ring isomorphism.

3.

+	O	I	J	K
O	O	I	J	K
I	I	O	K	J
J	J	K	O	I
K	K	J	I	O

×	O	I	J	K
O	O	O	O	O
I	O	I	J	K
J	O	J	O	J
K	O	K	J	I

$\{O, I, J, K\}$ is not a field, as J has no multiplicative inverse.

+	O	I	L	M
O	O	I	L	M
I	I	O	M	L
L	J	M	O	I
M	M	L	I	O

×	O	I	L	M
O	O	O	O	O
I	O	I	L	M
L	O	L	M	I
M	O	M	I	L

$\{O, I, L, M\}$ is a field. This is one way of constructing $\mathbb{F}_4$.

4. $\|u\|^2 = 4 + 9 + 16 + 1 = 30$ so $u^{-1} = \frac{1}{30}(2 + 3i - 4j - k)$.

$\|v\|^2 = 1 + 1 + 4 + 9 = 15$ so $v^{-1} = \frac{1}{15}(-1 - i - 2j + 3k)$.

5. Let S be the given subset of $\mathbb{H}$. Then $0 = 0 + 0i \in S$ so S is non-empty.

Let $a + bi, c + di \in S$. Then $(a + bi) - (c + di) = (a - c) + (b - d)i \in S$

and $(a + bi)(c + di) = (ac - bd) + (ad + bc)i \in S$, so S is a subring of $\mathbb{H}$.

It is commutative (in fact it is a field, isomorphic to $\mathbb{C}$).

6. Let $u = a + bi + cj + dk$, $v = e + fi + gj + hk$, so

$$\psi(u + v) = \begin{pmatrix} (a + e) + (b + f)i & -(c + g) - (d + h)i \\ (c + g) - (d + h)i & (a + e) - (b + f)i \end{pmatrix}$$

$$= \begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix} + \begin{pmatrix} e + fi & -g - hi \\ g - hi & e - fi \end{pmatrix} = \psi(u) + \psi(v).$$

$uv = (ae - bf - cg - dh) + (af + be + ch - dg)i + (ag - bh + ce + df)j + (ah + bg - cf + de)k$, so $\psi(uv) =$

$$\begin{pmatrix} (ae - bf - cg - dh) + (af + be + ch - dg)i & -(ag - bh + ce + df) - (ah + bg - cf + de)i \\ (ag - bh + ce + df) - (ah + bg - cf + de)i & (ae - bf - cg - dh) - (af + be + ch - dg)i \end{pmatrix}.$$

$\psi(u)\psi(v) = \begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix} \begin{pmatrix} e + fi & -g - hi \\ g - hi & e - fi \end{pmatrix}$, which expands to give the same matrix as $\psi(uv)$, so ψ is a ring homomorphism.

Suppose $\psi(u) = 0$. Then as the set $\{1, i\}$ is linearly independent over $\mathbb{R}$, $a = b = c = d = 0$, so $u = 0$. Thus $\text{Ker } \psi = \{0\}$, so ψ is injective.

7. $u = a + bi + cj + dij = a + bi + (c + di)j$, so $u = v + wj$ where $v = a + bi$, $w = c + di$.

$$\bar{u} = a - bi - cj - dij = a - bi - cj + dji = a - bi - j(c - di) = \bar{v} - j\bar{w}.$$

8. $u\bar{u} = (a + bi + cj + dk)(a - bi - cj - dk) = a^2 - abi - acj - adk + bai - b^2i^2 - bcij - bdk - cbj - c^2j^2 - cdjk + dak - dbki - dckj - d^2k^2$

Using $ji = -ij$, $ki = -ik$, $kj = -jk$ and $i^2 = j^2 = k^2 = -1$, the expression simplifies to $a^2 + b^2 + c^2 + d^2$.

Let $v = p + qi + rj + sk$. Then $\bar{iv} = \overline{-q + pi - sj + rk} = -q - pi + sj - rk$

$$= (p + qi + rj + sk)(-i) = \bar{v}\bar{i} \text{ and similarly } \bar{jv} = \bar{v}\bar{j} \text{ and } \bar{kv} = \bar{v}\bar{k}.$$

$$\text{Hence } \overline{uv} = \overline{av + biv + cjv + dkv} = \bar{a}\bar{v} + \bar{b}\bar{iv} + \bar{c}\bar{jv} + \bar{d}\bar{kv}$$

$$= \bar{a}\bar{v} + \bar{b}\bar{v}\bar{i} + \bar{c}\bar{v}\bar{j} + \bar{d}\bar{v}\bar{k}$$

$$= \bar{v}(a - bi - cj - dk) = \bar{v}\bar{u}.$$

It follows that $\|uv\|^2 = uv\overline{uv} = uv\bar{v}\bar{u} = u\|v\|^2\bar{u} = \|v\|^2u\bar{u} = \|v\|^2\|u\|^2 = \|u\|^2\|v\|^2$, so $\|uv\| = \|u\| \cdot \|v\|$.

9. Let $u = 2 + 4i + 6j + 8k$, $v = 3 + 5i + 7j + 9k$, or other suitable quaternions.

Then $uv = -128 + 20i + 36j + 40k$, so $\{w, x, y, z\} = \{20, 36, 40, 128\}$ (not unique).